📘 Project Logbook

Week 5: ARCore Integration and Environment Scanning

Overview:

This week was centered on AR functionality — allowing the app to detect surfaces and place virtual markers. The team experimented with ARCore to understand how users could view facilities through their device’s camera in a real-world environment.

Progress:

- Integrated ARCore and Sceneform SDKs into the Android project.
- Activated camera permissions and set up AR scene with surface detection.
- Developed scripts to detect horizontal planes and place virtual anchors.
- Rendered 3D models and arrows pointing to mock facilities like “Toilet,” “Platform 2,” and “Exit.”
- Connected virtual anchors to database-stored coordinates for consistency.
- Built a feature to reload and retrieve anchors based on location queries.
- Added debugging tools for anchor position accuracy and depth adjustments.
- Simulated real-world environments using objects and floor markers to mimic station layout.

Summary:

The app now supports real-time AR marker placement. While tests were run in a controlled environment, this provided valuable insight into how spatial data will translate visually in stations.

Next Week and Beyond:

Next week involves implementing the A* pathfinding algorithm to compute efficient navigation routes and overlay them in AR with directional graphics.